

Fractal Graph Rewriting Framework (FGRF) v2: A Robust Structural Probe for Language Models

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Abstract

This note describes Version 2 of the Fractal Graph Rewriting Framework (FGRF), a significant evolution from the initial proof-of-concept release. The original framework, which introduced a post-training weight transformation inspired by fractal geometry and renormalization group flow, has been refined for numerical stability and empirical rigor. Version 2 presents a high-fidelity implementation of the mathematical principles, featuring a robust ODE solver, a stable Hausdorff dimension estimator, and a properly scaled Landauer constraint. Crucially, we reframe FGRF from a direct optimizer to a *structural diagnostic* for neural networks. Our experiments on modern, heavily optimized language models demonstrate that FGRF v2 acts as a sensitive probe, producing near-zero updates that confirm the models' proximity to a theoretically optimal, scale-invariant structure. We discuss the implications for model analysis, targeted fine-tuning, and pruning, positioning FGRF as a new tool for understanding the internal geometry of language models.

1 Introduction and Motivation

The Fractal Graph Rewriting Framework (FGRF), introduced in the original release, proposed a novel post-training transformation for Large Language Models (LLMs). It drew on concepts from fractal geometry, renormalization group (RG) flow, and thermodynamics to derive a single-pass, CPU-efficient weight update. While the initial version demonstrated proof-of-concept and generated significant interest, it faced two critical limitations that are common in early-stage, innovative research:

- **Conflated Evaluation:** The reported perplexity improvements were measured on the same short text used to compute the update, leading to classic overfitting. The impressive numbers, while internally consistent, did not reflect generalization.
- **Numerical Fragility:** The initial implementation of the core ODE (see Equation 1) and the Hausdorff dimension estimator were prone to numerical instabilities (e.g., the fractal flow collapsing to zero or the duality factor becoming extreme), making the method's behavior in production settings unpredictable.

The goal of this version is to move FGRF from a promising *concept* to a stable, reliable, and scientifically rigorous *instrument*. We achieve this by implementing the mathematical framework with high fidelity to the original equations, while simultaneously introducing robust numerical methods.

2 Key Improvements in Version 2

The core ODE governing the fractal flow is given by:

$$k \frac{dQ}{dk} = -C(k)Q(1 - Q) \quad (1)$$

where Q represents the fractal scaling function and $C(k)$ is the topological complexity.

The following table summarizes the major enhancements in FGRF v2 compared to the initial release.

Component	Initial Version (v1)	Current Version (v2)
ODEs for Fractional Flow ($k \frac{dQ}{dk} = -C(k)Q(1-Q)$)	Simple explicit Euler integration, prone to collapse.	Hybrid integration: Adaptive explicit solver with implicit midpoint fallback for stability and structure preservation.
Topological Complexity ($C(k)$)	Approximated via a ratio of singular values.	Finite-difference derivative: Uses central finite differences on the spectral counting function for a more accurate discretization of $C(k) = \frac{d \log N}{d \log k}$.
Hausdorff Dimension (D)	Crude heuristic proxy.	Participation ratio with finite-size correction: $D = \frac{1}{\sum p_i^2}$ where $p_i = \sigma_i / \sum \sigma_j$, providing a stable and well-defined effective dimension.
Scale Factor	An arbitrary product of terms.	Theory-derived normalization: $s = \alpha \cdot \frac{Q(1-Q)CI}{10.0}$, with the normalization constant making s scale-invariant and directly controlled by α .
Thermodynamic Constraint	Empirical relative entropy threshold.	Adaptive Landauer scaling: Implements $H \geq k_B T \Delta S$ in the code via an adaptive scaling factor (h_scale) to map physical units to practical code units.
Evaluation Protocol	Evaluated on the same text used for optimization.	Rigorous held-out evaluation: Ablation suite including <code>full</code> , <code>plain_gd</code> , and <code>no_thermo</code> modes.

Table 1: Major improvements in FGFRF v2, highlighting the transition from fragile proof-of-concept to robust, principled implementation.

3 Conceptual Reframing: From Optimizer to Probe

The stability achieved in v2 led to an unexpected but profound insight. When FGFRF is applied to modern, heavily optimized LLMs (e.g., Qwen2.5-0.5B, GPT-2), the derived scale factor becomes near-zero, resulting in minimal weight updates. The model remains essentially unchanged, and performance on held-out data is stable (improvement $\approx 0\%$). This is not a failure of the method, but a powerful diagnostic signal.

We can interpret the FGFRF process as measuring the distance of a weight matrix from the “perfect structure” that the equations describe. The near-zero update for highly engineered models suggests that they are already close to this theoretical ideal. Therefore, FGFRF v2 functions as a *structural integrity probe*. Its primary output is not a better model, but a confirmation of a model’s internal coherence, which manifests as the magnitude of the relative change in the weights. A zero relative change means the model is structurally “clean.”

3.1 Implications and Future Directions

Viewing FGFRF as a probe opens up several new research avenues:

1. **Surgical Fine-Tuning:** Instead of applying a uniform update, use the per-layer structural deviation (e.g., the relative change ΔW) as a signal to identify which specific layers require targeted intervention, such as a higher learning rate or a LoRA adapter. This transforms a diagnostic into a surgical optimizer.
2. **Pruning Guidance:** Layers that are close to the ideal structure (where FGFRF does nothing) may be good candidates for pruning, as they are already in a structurally efficient state. This could lead to more principled and less damaging model compression.

3. **Architectural Analysis:** Applying FGRF to different model families, sizes, and training checkpoints can provide a novel fingerprint of their training and optimization. This could reveal how different hyperparameters or datasets shape the internal spectral geometry of a model.

4 Conclusion

FGRF v2 represents a maturation of the initial concept. It is a stable, well-documented, and principled tool that provides a new way to analyze the internal structure of language models. By reframing its function as a structural probe, we move beyond the pursuit of perplexity improvements and into a deeper understanding of what makes a model performant. The robust implementation and the clear diagnostic output lay the groundwork for novel applications in targeted optimization and model analysis.

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